```
# Importing pandas
import pandas as pd
```

```
# Importing Dataset
data=pd.read_csv("/content/Global Dataset Task 2.csv")
```

```
# First five rows
print("Starting data:",data.head())
```

```
Starting data:           Category         City        Country Customer.ID     Customer.Name  \
0  Office Supplies  Los Angeles  United States   LS-172304  Lycoris Saunders
1  Office Supplies  Los Angeles  United States   MV-174854     Mark Van Huff
2  Office Supplies  Los Angeles  United States   CS-121304      Chad Sievert
3  Office Supplies  Los Angeles  United States   CS-121304      Chad Sievert
4  Office Supplies  Los Angeles  United States   AP-109154    Arthur Prichep

   Discount Market  记录数              Order.Date        Order.ID  ... Sales  \
0       0.0     US    1  2011-01-07 00:00:00.000  CA-2011-130813  ...    19
1       0.0     US    1  2011-01-21 00:00:00.000  CA-2011-148614  ...    19
2       0.0     US    1  2011-08-05 00:00:00.000  CA-2011-118962  ...    21
3       0.0     US    1  2011-08-05 00:00:00.000  CA-2011-118962  ...   111
4       0.0     US    1  2011-09-29 00:00:00.000  CA-2011-146969  ...     6

    Segment                Ship.Date       Ship.Mode  Shipping.Cost  \
0  Consumer  2011-01-09 00:00:00.000    Second Class           4.37
1  Consumer  2011-01-26 00:00:00.000  Standard Class           0.94
2  Consumer  2011-08-09 00:00:00.000  Standard Class           1.81
3  Consumer  2011-08-09 00:00:00.000  Standard Class           4.59
4  Consumer  2011-10-03 00:00:00.000  Standard Class           1.32

        State  Sub.Category  Year        Market2 weeknum
0  California         Paper  2011  North America       2
1  California         Paper  2011  North America       4
2  California         Paper  2011  North America      32
3  California         Paper  2011  North America      32
4  California         Paper  2011  North America      40

[5 rows x 27 columns]
```

```
# Last five rows
print(data.tail())
```

```
             Category         City        Country Customer.ID  \
51285  Office Supplies  Los Angeles  United States   AM-103604
51286  Office Supplies  Los Angeles  United States   AM-103604
51287  Office Supplies  Los Angeles  United States   HR-147704
51288  Office Supplies  Los Angeles  United States   RM-196754
51289  Office Supplies  Los Angeles  United States   FH-143654

        Customer.Name  Discount Market  记录数              Order.Date  \
51285  Alice McCarthy       0.2     US    1  2014-12-03 00:00:00.000
51286  Alice McCarthy       0.2     US    1  2014-12-03 00:00:00.000
51287  Hallie Redmond       0.2     US    1  2014-12-18 00:00:00.000
51288   Robert Marley       0.2     US    1  2014-12-25 00:00:00.000
51289    Fred Hopkins       0.2     US    1  2014-12-26 00:00:00.000

             Order.ID  ... Sales      Segment                Ship.Date  \
51285  CA-2014-109701  ...    69    Corporate  2014-12-04 00:00:00.000
51286  CA-2014-109701  ...     9    Corporate  2014-12-04 00:00:00.000
51287  CA-2014-106964  ...    12  Home Office  2014-12-21 00:00:00.000
51288  CA-2014-145219  ...    90  Home Office  2014-12-26 00:00:00.000
51289  CA-2014-121398  ...   154    Corporate  2014-12-30 00:00:00.000

            Ship.Mode  Shipping.Cost       State  Sub.Category  Year  \
51285        Same Day           5.15  California       Binders  2014
51286        Same Day           0.44  California       Binders  2014
51287    First Class            0.31  California       Binders  2014
51288    First Class           15.95  California       Binders  2014
51289  Standard Class           9.59  California       Binders  2014

            Market2 weeknum
51285  North America      49
51286  North America      49
51287  North America      51
51288  North America      52
51289  North America      52

[5 rows x 27 columns]
```

```
# Basic Infromation
print(data.info)
```

```
51286  Office Supplies  Los Angeles  United States   AM-103604
51287  Office Supplies  Los Angeles  United States   HR-147704
51288  Office Supplies  Los Angeles  United States   RM-196754
51289  Office Supplies  Los Angeles  United States   FH-143654

          Customer.Name  Discount Market  记录数              Order.Date  \
0      Lycoris Saunders       0.0     US    1  2011-01-07 00:00:00.000
1         Mark Van Huff       0.0     US    1  2011-01-21 00:00:00.000
2          Chad Sievert       0.0     US    1  2011-08-05 00:00:00.000
3          Chad Sievert       0.0     US    1  2011-08-05 00:00:00.000
4        Arthur Prichep       0.0     US    1  2011-09-29 00:00:00.000
...                 ...       ...    ...  ...                      ...
51285    Alice McCarthy       0.2     US    1  2014-12-03 00:00:00.000
51286    Alice McCarthy       0.2     US    1  2014-12-03 00:00:00.000
51287    Hallie Redmond       0.2     US    1  2014-12-18 00:00:00.000
51288     Robert Marley       0.2     US    1  2014-12-25 00:00:00.000
51289      Fred Hopkins       0.2     US    1  2014-12-26 00:00:00.000

             Order.ID  ... Sales      Segment               Ship.Date  \
0      CA-2011-130813  ...    19     Consumer  2011-01-09 00:00:00.000
1      CA-2011-148614  ...    19     Consumer  2011-01-26 00:00:00.000
2      CA-2011-118962  ...    21     Consumer  2011-08-09 00:00:00.000
3      CA-2011-118962  ...   111     Consumer  2011-08-09 00:00:00.000
4      CA-2011-146969  ...     6     Consumer  2011-10-03 00:00:00.000
...               ...  ...   ...          ...                      ...
51285  CA-2014-109701  ...    69    Corporate  2014-12-04 00:00:00.000
51286  CA-2014-109701  ...     9    Corporate  2014-12-04 00:00:00.000
51287  CA-2014-106964  ...    12  Home Office  2014-12-21 00:00:00.000
51288  CA-2014-145219  ...    90  Home Office  2014-12-26 00:00:00.000
51289  CA-2014-121398  ...   154    Corporate  2014-12-30 00:00:00.000

            Ship.Mode  Shipping.Cost       State  Sub.Category  Year  \
0        Second Class           4.37  California         Paper  2011
1      Standard Class           0.94  California         Paper  2011
2      Standard Class           1.81  California         Paper  2011
3      Standard Class           4.59  California         Paper  2011
4      Standard Class           1.32  California         Paper  2011
...               ...            ...         ...           ...   ...
51285       Same Day           5.15  California       Binders  2014
51286       Same Day           0.44  California       Binders  2014
51287     First Class           0.31  California       Binders  2014
51288     First Class          15.95  California       Binders  2014
51289  Standard Class           9.59  California       Binders  2014

             Market2 weeknum
0      North America       2
1      North America       4
2      North America      32
3      North America      32
4      North America      40
...              ...     ...
51285  North America      49
51286  North America      49
51287  North America      51
51288  North America      52
51289  North America      52

[51290 rows x 27 columns]>
```

```
# Total Rows and columns
print(data.shape)
```

```
(51290, 27)
```

```
# Columns names
print("\n Column Name:")
print(data.columns.to_list())
```

```

 Column Name:
['Category', 'City', 'Country', 'Customer.ID', 'Customer.Name', 'Discount', 'Market', '记录数', 'Order.Date', 'Order.ID', 'Order
```

```
# Summary
print(data.describe())
```

```
           Discount      记录数        Profit      Quantity       Row.ID  \
count  51290.000000  51290.0  51290.000000  51290.000000  51290.00000
```

```
mean       0.142908      1.0     28.610982      3.476545  25645.50000
std        0.212280      0.0    174.340972      2.278766  14806.29199
min        0.000000      1.0  -6599.978000      1.000000      1.00000
25%        0.000000      1.0      0.000000      2.000000  12823.25000
50%        0.000000      1.0      9.240000      3.000000  25645.50000
75%        0.200000      1.0     36.810000      5.000000  38467.75000
max        0.850000      1.0   8399.976000     14.000000  51290.00000

              Sales  Shipping.Cost          Year       weeknum
count  51290.000000   51290.000000  51290.000000  51290.000000
mean     246.498440      26.375818   2012.777208     31.287112
std      487.567175      57.296810      1.098931     14.429795
min        0.000000       0.002000   2011.000000      1.000000
25%       31.000000       2.610000   2012.000000     20.000000
50%       85.000000       7.790000   2013.000000     33.000000
75%      251.000000      24.450000   2014.000000     44.000000
max    22638.000000     933.570000   2014.000000     53.000000
```

```
# Finding missing values
print("\nMissing Values in Each Column:")
print("\n",data.isnull().sum())
```

```
Missing Values in Each Column:

 Category          0
City               0
Country            0
Customer.ID        0
Customer.Name      0
Discount           0
Market             0
记录数                0
Order.Date         0
Order.ID           0
Order.Priority     0
Product.ID         0
Product.Name       0
Profit             0
Quantity           0
Region             0
Row.ID             0
Sales              0
Segment            0
Ship.Date          0
Ship.Mode          0
Shipping.Cost      0
State              0
Sub.Category       0
Year               0
Market2            0
weeknum            0
dtype: int64
```

```
print(data.dropna(inplace=True))
```

```
None
```

```
# Duplicate data
print("Total duplicate rows:",data.duplicated().sum())
```

```
Total duplicate rows: 0
```

```
# Removing duplicates
data=data.drop_duplicates()
```

```
# Datatype
print("\nData Types and Missing Values:")
print(data.dtypes)
```

```
Data Types and Missing Values:
Category           object
City               object
Country            object
Customer.ID        object
Customer.Name      object
Discount          float64
Market             object
```

```
记录数                int64
Order.Date         object
Order.ID           object
Order.Priority     object
Product.ID         object
Product.Name       object
Profit            float64
Quantity            int64
Region             object
Row.ID              int64
Sales               int64
Segment            object
Ship.Date          object
Ship.Mode          object
Shipping.Cost     float64
State              object
Sub.Category       object
Year                int64
Market2            object
weeknum             int64
dtype: object
```

```
print(data.dtypes)
```

```
Category           object
City               object
Country            object
Customer.ID        object
Customer.Name      object
Discount          float64
Market             object
记录数                int64
Order.Date         object
Order.ID           object
Order.Priority     object
Product.ID         object
Product.Name       object
Profit            float64
Quantity            int64
Region             object
Row.ID              int64
Sales               int64
Segment            object
Ship.Date          object
Ship.Mode          object
Shipping.Cost     float64
State              object
Sub.Category       object
Year                int64
Market2            object
weeknum             int64
dtype: object
```

```
data['Order.Date']
```

	Order.Date
0	2011-01-07
1	2011-01-21
2	2011-08-05
3	2011-08-05
4	2011-09-29
...	...
51285	2014-12-03
51286	2014-12-03
51287	2014-12-18
51288	2014-12-25
51289	2014-12-26

51290 rows × 1 columns

dtype: datetime64[ns]

```
data['Order.Date']=pd.to_datetime(data['Order.Date'])
```

```
data['Order.Date']
```

	Order.Date
0	2011-01-07
1	2011-01-21
2	2011-08-05
3	2011-08-05
4	2011-09-29
...	...
51285	2014-12-03
51286	2014-12-03
51287	2014-12-18
51288	2014-12-25
51289	2014-12-26

51290 rows × 1 columns

dtype: datetime64[ns]

```
data['Ship.Date']
```

	Ship.Date
0	2011-01-09 00:00:00.000
1	2011-01-26 00:00:00.000
2	2011-08-09 00:00:00.000
3	2011-08-09 00:00:00.000
4	2011-10-03 00:00:00.000
...	...
51285	2014-12-04 00:00:00.000
51286	2014-12-04 00:00:00.000
51287	2014-12-21 00:00:00.000
51288	2014-12-26 00:00:00.000
51289	2014-12-30 00:00:00.000

51290 rows × 1 columns

dtype: object

```
data['Ship.Date']=pd.to_datetime(data['Ship.Date'])
```

```
data['Ship.Date']
```

	Ship.Date
0	2011-01-09
1	2011-01-26
2	2011-08-09
3	2011-08-09
4	2011-10-03
...	...
51285	2014-12-04
51286	2014-12-04
51287	2014-12-21
51288	2014-12-26
51289	2014-12-30

51290 rows × 1 columns

dtype: datetime64[ns]

```
data[['Order.Date','Ship.Date']].dtypes
```

	0
Order.Date	datetime64[ns]
Ship.Date	datetime64[ns]

dtype: object

```
data.to_csv("Global_Sales_Cleaned_Dataset.csv",index=False)
```

```
from google.colab import files
```

```
files.download("Global_Sales_Cleaned_Dataset.csv")
```

```
Start coding or generate with AI.
```